

# Thomas Moerman



PhD researcher in natural language processing and AI, Ghent University (LT3)

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My research develops retrieval-augmented and synthetic-data methods for language technology, with machine translation as the primary focus and applications in educational NLP, fairness in language models, and vision-language OCR. A consistent finding is that these techniques let smaller specialized models reach or exceed the quality of much larger general-purpose LLMs.

## EDUCATION

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| <b>PhD in Natural Language Processing</b> , Ghent University, LT3<br>Machine translation augmented with automatically extracted similar translations. Supervisors: Arda Tezcan and Els Lefever. | 2023 to 2027 (expected) |
| <b>Advanced MA in Linguistics: Natural Language Processing</b> , Ghent University and KU Leuven, summa cum laude                                                                                | 2023                    |
| <b>MA in Linguistics and Literature</b> , English and Scandinavian studies, Ghent University, cum laude                                                                                         | 2022                    |
| <b>BA in Linguistics and Literature</b> , English and Swedish, Ghent University, cum laude                                                                                                      | 2021                    |

## SELECTED PUBLICATIONS

9 published or accepted works, 6 first-authored, 3 journal articles. Full list: [Google Scholar](#) and [ORCID](#)

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| Moerman, Gkovedarou, Hackenbuchner. <b>ContraGAND: Auditing and Repairing Gender Ambiguity Failures in LLMs with Neurosymbolic Contrastive Data Augmentation</b> . EMNLP, main conference. | 2026 |
| Moerman, Tezcan, Macken. <b>Multilingual Communication in the Asylum Context: LLM-Based MT with Fuzzy Match Augmentation</b> . EAMT. Shortlisted for the best-paper award.                 | 2026 |
| Moerman, Degraeuwe, Tezcan. <b>Fuzzy Semantic Retrieval Strategies for Automated Short-Answer Grading with LLMs</b> . Computational Linguistics in the Netherlands Journal, 15.            | 2026 |
| Moerman, Tezcan. <b>Getting More from Small LLMs: Fuzzy-Match Retrieval and Candidate Scoring for Multitask Sorbian NLP</b> . WMT shared task system description, under review.            | 2026 |
| Moerman, Lefever, Tezcan. <b>Advancing Fuzzy Match Augmentation for Domain-Specific Machine Translation</b> . Under review at the Journal of Artificial Intelligence Research.             | 2025 |

## AWARDS AND COMPETITIONS

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Winner, Sorbian track, WMT 2026 shared task on multitask LLMs with limited resources</b> (team LT3, team lead). One 2B model jointly handling translation, QA, spelling, grammar, and maths tasks for Upper and Lower Sorbian. | 2026 |
| <b>Best-paper award shortlist</b> , EAMT 2026.                                                                                                                                                                                    | 2026 |

## RESEARCH GRANTS

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Principal applicant</b> , VSC Tier-1 compute project grant on the Sofia cluster (38,546 GPU hours), after earlier Tier-1 project allocations (2024, 2025) and a Tier-1 Starting Grant (2024). | 2024 to 2026 |
| <b>Named collaborator</b> on five further Tier-1 compute grants across four Ghent University research groups, contributing LLM fine-tuning and retrieval pipelines.                              | 2025 to 2026 |

## TEACHING

Guest lectures on Neural Machine Translation and on LLMs in Translation (MTPE course, Ghent University). Co-taught Introduction to Python for Humanities. Hands-on HPC workshop for the LT3 team. Python module at the MILS summer school (Methods in Language Sciences).

## ACADEMIC SERVICE

Reviewer for ARR, EAMT, MT Summit, and CLIN (2023, 2025). Presented at EAMT 2026, MT Summit 2025, the Machine Translation Marathon 2025, CLIN, and LREC 2026.

## TECHNICAL SKILLS AND LANGUAGES

Python, PyTorch, Hugging Face Transformers, vLLM, axolotl, LoRA fine-tuning, FAISS retrieval, OpenNMT, Moses, SLURM on Tier-1 HPC (A100 and H200 clusters).  
Dutch (native), English (C2), Swedish (C1), French (B2), German (B1), Norwegian (B1).